

LONDON STANSTED, United Kingdom

WMO: 036830

Lat: 51.885N

Lon: 0.235E

Elev: 348

StdP: 14.51

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.0	28.0	20.9	15.9	29.0	23.1	17.6	30.5	30.0	48.3	27.3	48.3	6.4	350	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	81.0	65.3	77.2	63.9	73.7	62.3	67.2	77.3	65.4	74.3	63.8	71.5	9.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	90.3	70.7	62.4	85.4	68.9	60.8	80.7	67.0	31.8	78.4	30.4	74.4	29.3	71.7	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.7	19.2	DB	20.0	87.7	4.5	3.4	16.8	90.2	14.1	92.2	11.6	94.1	8.3	96.6	(4)
(5)			WB	19.4	70.7	4.5	2.0	16.2	72.1	13.6	73.2	11.1	74.3	7.8	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	39.5	40.2	43.4	47.8	53.3	58.8	63.2	62.8	57.9	51.9	44.8	40.4
	DBStd	10.08	5.77	5.89	5.49	5.59	5.52	4.96	4.92	4.79	4.62	5.33	5.61	6.49
	HDD50	1469	328	277	211	106	29	1	0	0	2	41	172	302
	HDD65	5427	792	695	668	516	365	196	96	103	218	408	607	763
	CDD50	1605	1	2	7	39	130	264	408	396	239	99	15	5
	CDD65	89	0	0	0	0	1	9	40	33	5	1	0	0
	CDH74	764	0	0	0	2	11	101	346	265	35	4	0	0
	CDH80	160	0	0	0	0	1	16	75	65	3	0	0	0
Wind	WSAvg	9.6	11.0	10.9	10.5	9.5	9.3	8.8	8.5	8.6	8.8	9.7	9.6	10.4
Precipitation	PrecAvg	24.5	2.2	1.6	1.8	1.8	1.8	1.9	2.0	2.1	2.3	2.3	2.4	2.3
	PrecMax	31.0	5.0	3.2	3.9	4.2	3.4	5.0	4.3	4.1	5.2	6.1	6.7	4.7
	PrecMin	18.1	0.4	0.5	0.2	0.3	0.2	0.1	0.2	0.1	0.2	0.1	0.8	0.6
	PrecStd	3.5	1.1	0.8	1.0	0.9	0.9	1.2	1.0	1.1	1.5	1.6	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	57.2	62.2	71.5	75.4	82.3	86.3	86.1	78.7	69.8	60.4	56.8
		MCWB	51.5	52.0	52.4	58.4	62.2	65.4	67.2	67.0	63.7	62.2	57.6	53.5
	2%	DB	52.2	53.4	57.5	64.8	71.6	76.8	81.7	80.9	73.1	64.5	57.4	54.0
		MCWB	49.8	49.9	50.6	54.5	60.3	63.6	65.5	65.3	62.2	59.0	54.8	51.6
	5%	DB	50.3	51.4	55.1	61.0	67.9	72.5	77.3	75.6	69.5	62.2	55.5	52.6
		MCWB	48.3	48.2	49.6	52.7	58.3	61.5	63.9	63.5	60.8	57.8	52.8	50.3
	10%	DB	48.4	48.8	52.1	57.5	64.2	69.4	73.8	72.0	66.3	59.8	53.5	50.3
		MCWB	46.8	46.5	47.6	50.6	56.0	60.1	62.6	61.6	58.7	56.4	51.0	48.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	52.9	54.3	59.3	65.0	68.4	69.9	69.3	66.9	62.8	58.1	54.5
		MCDB	54.2	56.3	59.3	69.3	72.4	79.1	82.6	82.2	74.5	68.5	59.8	56.3
	2%	WB	51.0	50.4	52.1	55.9	61.4	64.9	67.1	66.7	64.1	60.3	55.6	52.8
		MCDB	52.3	52.4	56.2	63.2	69.0	73.0	77.5	77.0	70.4	63.4	57.4	54.6
	5%	WB	48.7	48.6	50.5	53.7	59.1	62.9	65.4	65.0	62.1	58.9	53.5	51.0
		MCDB	50.0	50.4	53.8	60.0	66.3	70.5	74.9	73.6	66.9	61.3	55.4	52.7
	10%	WB	46.8	46.8	48.7	51.6	57.0	60.9	63.7	63.3	60.3	57.2	51.5	48.7
		MCDB	48.2	48.7	51.8	56.7	62.9	67.6	71.9	70.4	64.7	59.6	53.3	50.1
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.5	12.9	15.9	16.4	16.7	17.6	16.9	15.2	12.6	10.2	8.9
		MCDBR	10.8	12.3	16.7	20.8	22.4	22.7	23.5	22.1	19.1	13.7	10.7	9.4
		MCWBR	10.4	9.6	10.9	12.1	12.8	11.7	10.1	9.5	10.0	9.3	9.3	9.1
	5% WB	MCDBR	10.8	11.1	14.8	18.9	20.1	20.0	21.0	19.3	16.0	12.3	10.2	9.5
		MCWBR	10.8	9.7	10.8	11.8	12.4	11.3	10.3	9.4	9.6	9.5	9.3	9.4
Clear-Sky Solar Irradiance	taub	0.324	0.340	0.375	0.397	0.402	0.405	0.402	0.392	0.372	0.356	0.328	0.314	
	taud	2.377	2.387	2.282	2.238	2.261	2.278	2.295	2.346	2.400	2.433	2.425	2.364	
	Ebn at Noon	214	246	257	266	270	270	268	265	257	237	212	197	
	Edh at Noon	19	25	34	39	40	40	39	35	30	24	18	17	
All-Sky Solar Radiation	RadAvg	244	457	808	1275	1535	1676	1595	1331	1000	578	297	192	
	RadStd	22	65	99	131	109	140	154	101	82	53	33	24	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A
(47) Regional (82 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page